

A Physics-Informed Data-Driven Friction Prediction Method for Crystal Growth Equipment

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Abstract—The nonlinear friction has a critical influence on the quality of single-crystal silicon (SCS), because of the lifting speed jitter and creeping phenomenon induced by the friction during the Czochralski crystal growth process. In this article, a new friction prediction method is proposed by fusing the improved LuGre friction model, lifting servo dynamic model, and measured data. The friction prediction physical information neural network (FP-PINN) is constructed, where the loss function consists of both physical information loss and data loss. The physical information loss includes the improved LuGre (ILG) model and servo dynamics. Where, ILG is modified by considering the crystal growth characteristic, i.e., the influence of SCS weight variation and rotation. In addition, a partial differential equation is constructed to describe behavior of the crystal weight variation. A weight estimation (WE)-PINN is then designed to solve the crystal weight, since the SCS weight variation caused by crystal growth and speed jitter is hard to calculate in real operation process. Finally, the obtained SCS weight is integrated into the FP-PINN to achieve an accurate and robust friction torque. Experiments are conducted to verify the accuracy and generalization ability of the proposed method.

Index Terms—Creeping phenomenon, friction prediction, lifting servo system, LuGre model, physical information neural network (PINN), single crystal silicon.

I. INTRODUCTION

SINGLE-CRYSTAL silicon (SCS) is crucial in the field of semiconductor industry and is widely used as the basic material for the manufacturing of integrated circuit chips [1], [2]. With the increasing of chip density, the requirements for the quality of SCS are raising. Czochralski (Cz) crystal growth

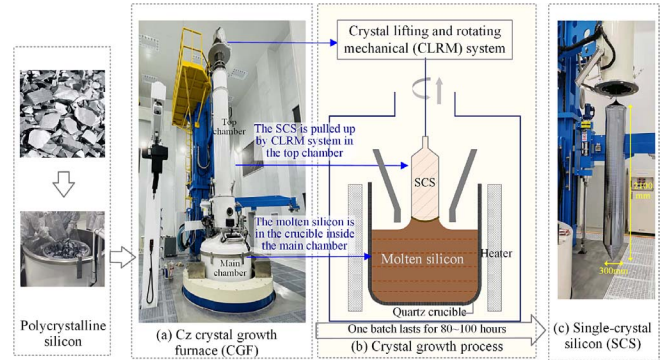


Fig. 1. Czochralski crystal growth process.

method is the mainstream approach currently to produce SCS [3], [4]. In Cz crystal growth process, the SCS is produced by pulling the crystal rod from the molten silicon in a vacuum-sealed furnace, as illustrated in Fig. 1.

The device that tailor-made for completing the SCS pulling motion together with rotation action in Cz crystal growth process can be called crystal lifting and rotating mechanical (CLRM) system [5]. The motion performance of CLRM is one of the vital factor that affects the quality of the produced SCS. The reason lies in the following aspects: 1) the action of SCS lifting and rotating is carried out throughout the process lasting for around hundred hours per batch. Any unstable action might cause that the silicon atoms cannot be arranged in a regular order, resulting in a decline in microscopic quality; and 2) the SCS lifting speed is ultra-low during the whole process. The friction effect on the lifting motion become greater, leading to speed jitter and the creep phenomenon of the SCS lifting. Considering the above factors, hence, the study of the friction in SCS lifting servo system is an important topic to maintain the stable motion during crystal growth, which is a needed for high-quality SCS preparation.

Currently, multiple methods have been proposed to describe friction characteristics, mainly including two categories: physical-based and learning-based model.

For the physical-based method, the LuGre model has been widely used in servo system friction modeling [6]. It is able to effectively capture the viscous, Coulomb friction, and dynamic hysteresis phenomena of friction, and is especially suitable for systems with complex nonlinear friction characteristics [7]. Currently, various improved LuGre models have been proposed.

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In [8], an improved LuGre is proposed to describe the friction characteristics between position, velocity, and direction. In [9], a point contact multibody model is constructed using 2-D LuGre friction model, based on which the dynamics of wheel mechanism is simulated. The stick-slip motions of the wheel are then analyzed. In [10], an improved LuGre model is proposed based on the microconvex assumption, Hertz contact theory, and thermal conditions. In [11], an improved LuGre model is used to calculate the static real-time steering torque of rubber tracked chassis. The above existing LuGre model and its improved versions consider the effects of working conditions for the specific mechanical system. But, these models cannot directly used for the SCS lifting servo system, since special movement characteristics are encountered such as SCS weight change and rotation motion influence.

Traditionally, simple nonlinear fitting techniques or purely data-driven approaches such as regression or shallow neural networks have been employed to approximate friction dynamics [12]. However, these methods often struggle to capture the strong coupling between velocity, load variations, and rotational effects that characterize the SCS lifting servo system. In particular, they lack the ability to generalize under varying operating conditions and typically require extensive retraining when the system state changes.

In recent years, the rise of deep learning technology provides new ideas for friction modeling [13]. Learning-based methods, especially physical information neural networks (PINNs), have demonstrated excellent results in many industrial processes [14], [15], [16]. Their nonlinear learning capability and strong generalization ability give them a wide range of application potentials for modeling the dynamics of mechanical system [17], [18]. In [19], a deep Lagrangian network is proposed, which can efficiently learn dynamics of a mechanical system, while ensuring the consistence with physical properties. In [20], a PINN-based friction-inclusive dynamics modeling method is proposed for industrial robots. A hybrid learning strategy for robot dynamics and friction is designed, ensuring modeling accuracy while avoiding reliance on the measurement of joint torque. In [21], a PINN surrogate model is designed for parameter estimation of Li-ion battery. Nevertheless, these PINN-based methods and the specialized learning strategies have not yet designed for friction modeling in the field of crystal growth.

Despite their respective strengths, existing physical models and data-driven methods each have limitations when dealing with the friction modeling in SCS lifting servo system. The physical models cannot adequately handle time-varying loads and rotational effects, whereas data-driven approaches often neglect the fundamental physics of friction, resulting in poor generalization under varying operational conditions.

In this article, a physics-informed data-driven method is newly proposed for friction modeling of a SCS lifting servo system, with fusion of improved physical-based LuGre model, servo dynamics, measured data, a hybrid PINN learning strategy. The proposed method is validated on a SCS lifting servo experimental platform in our laboratory. The main contributions of this article include

- 1) An improved physical-based friction model is established. By introducing the characteristics of the crystal rod, i.e., time-varying weight and rotation movement, into the traditional LuGre model, the friction characteristics during the SCS lifting motion process can be described precisely. The dynamic simulation of velocity jitter and low-speed crawling phenomena is achieved during the ultra-low-speed pulling process.
- 2) A new friction prediction method is proposed by fusing the physical-based friction model, lifting servo dynamic model, and measured data. By integrating the experiment data, this method can robustly adapt to the actual crystal growth process with different working conditions.
- 3) A two-stage hybrid learning strategy is constructed, which contains a weight estimation (WE)-PINN module for SCS weight estimation and an friction prediction (FP)-PINN module for friction prediction. By the WE-PINN, the weight can be accurately achieved even with the interference of the creep phenomenon. The weight estimation is integrated into the FP-PINN to predict the friction accurately when the weight measurement is hard to be achieved due to the limitation caused by the structure of the crystal growth furnace.

Compared with traditional methods, by integrating an improved LuGre model with a hybrid PINN framework, the proposed method not only enforces physically meaningful constraints but also enables simultaneous learning of friction parameters and time-varying loads under ultra-low-speed conditions. This article is structured as follows. Section II introduces problem description and the overall structure of the article. Section III presents the physical model of the servo system and friction dynamics. In Section IV, the proposed FP-PINN and WE-PINN are described. The experimental validation is shown in Section V. Finally, Section VI summarizes the article.

II. PROBLEM DESCRIPTION

A. SCS Preparation by Cz Crystal Growth Method

SCS is an important material for integrated circuit chip. The mainstream production method of SCS is by melting the polycrystalline silicon in the crystal growth furnace (CGF) shown in Fig. 1(a) and pulling the crystal from molten silicon. That is the so-called Cz crystal growth method, as depicted in Fig. 1(b). To produce a SCS rod with diameter of 300 mm and length of around 2000 mm (shown in Fig. 1(c)), the growth process usually lasts for around 100 h.

A typical mechanical structure drawing of CGF is shown in Fig. 2(a), where the crystal lifting and rotation mechanical (CLRM) system is the key device for crystal motion, which works throughout the whole growth process. Fig. 2(b) presents the machine drawing of CLRM system. Its main transmission components include motor for crystal lifting, reducer, and wire wheel. As shown in Fig. 2(b), a motor and a precision reducer driving the soft shaft (Tungsten rope) to wrap around the wire reel. The rope, connected to the crystal rod (i.e., SCS rod) through a seed crystal clamp, facilitates the lifting of the SCS rod. Hence it can also be called *SCS lifting servo system*.

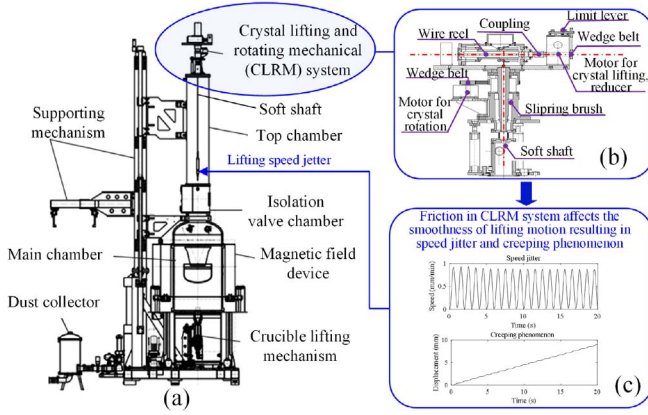


Fig. 2. Crystal growth equipment. (a) CGF. (b) CLRM. (c) Creeping phenomenon.

Another motor is used for the smooth rotation of the soft shaft, i.e., SCS rotation. The rotation and lifting are coupled to realize crystal movement.

B. Creeping Phenomenon in SCS Lifting Motion

In the above described crystal growth process, the SCS is pulled in ultra-low speed for long periods of time. The working time for growing one SCS rod is around 80–100 h. The lifting speed is usually in the range of [0.5, 2] mm/min. During this process, the SCS lifting motion can be affected by nonlinear friction, the accuracy of the optical encoder, and the system's coupling torque, resulting in irregular vibrations. Notably, nonlinear friction interference is a primary cause of speed jitter and low-speed creep [22], which can severely compromise the quality of SCS, as depicted in Fig. 2(c). In the context of SCS lifting motion, friction arises between the wire reel, coupling, and soft shaft.

Thus, establishing precise friction models and identifying model parameters under varying working conditions holds substantial theoretical and practical significance for high-quality crystal growth. Such models could aid in friction compensation and precision control of the motion, enhancing the quality of the produced SCS.

C. Overall Structure of the Proposed Method

Fig. 3 presents the structure of the proposed physics-informed data-driven friction prediction method. First, the experiment data is achieved from the constructed SCS lifting servo experimental platform as shown in Fig. 3(a).

Then, the SCS weight change behavior is depicted by a partial derivation equation (PDE), which is varied with the lifting speed and time. The SCS weight is calculated by a weight-estimation PINN method as shown in Fig. 3(b), which is designed with a combined loss function including both data loss and physical constraint loss.

As shown in Fig. 3(c), a friction-prediction PINN is then designed. Before that, a physical friction model is constructed based on the improved LuGre model with extra consideration

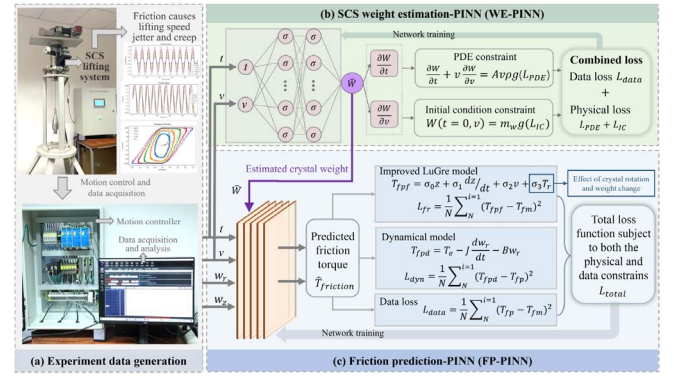


Fig. 3. Structure of proposed method.

given to the change of the SCS weight and the rotation of the SCS, which can well fit the actual operating characteristics of the SCS lifting servo system. The improved LuGre model and the servo dynamics model are integrated into the FP-PINN network as physical information terms. By combining the physical loss with data loss, a comprehensive friction modeling framework is constructed. This approach can describe the dynamical behavior of the friction more comprehensively and improve the modeling accuracy.

Finally, the WE-PINN and FP-PINN are working together to achieved the prediction of friction torque.

In following sections, the physical model including the lifting servo dynamics and the improved LuGre model are described first. Then the hybrid WE-PINN and FP-PINN learning strategy are presented.

III. PHYSICS-BASED MODEL

A. Dynamical Model of SCS Lifting Servo System

The dynamical behavior of the SCS lifting servo system can be described by the torque balance equation of the mechanical system. The main power source of the system is a permanent magnet synchronous motor (PMSM) [23], [24], [25], whose output torque is used to overcome the frictional and viscous resistance, and drive the load to generate acceleration. The dynamics equation is given by

$$J \frac{d\omega_r}{dt} = T_e - B\omega_r - T_f \quad (1)$$

where J denotes the equivalent rotational inertia of the system, ω_r is the angular velocity of the system, T_e is the electromagnetic torque, B is the viscous damping coefficient, and T_f is the frictional torque.

In this article, the focus is on the mechanical dynamic behavior of the lifting servo system with torque balance as the core, hence, the influence of electrical dynamics [26], [27] is omitted. In addition, the friction torque T_f is regarded as the main load torque. The model is able to reveal the mechanism of friction on the dynamics of lifting servo system. It provides the one of physical constraints as depicted in Fig. 4(a) for the following friction prediction.

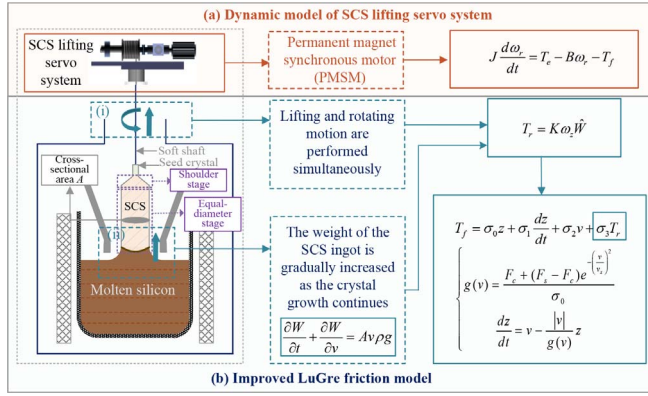


Fig. 4. SCS lifting dynamics and improved LuGre friction model.

B. Improved LuGre Friction Model

During the crystal growth process, the SCS lifting servo system runs for a long time in ultra-low-speed mode. The non-linear friction disturbances on the lifting servo system results in speed jitter and low-speed creep of SCS lifting motion, which seriously affect the quality of the produced SCS. Therefore, it is particularly important to construct a friction model that can accurately describe the friction characteristics of the system [28], [29]. The LuGre model is widely used in the dynamic modeling of friction because of its ability to effectively characterize the dip-slip phenomenon, friction hysteresis, and preslip displacement [30], [31]. The mathematical model of LuGre friction is described by

$$T_f = \sigma_0 z + \sigma_1 \frac{dz}{dt} + \sigma_2 v \quad (2)$$

where z is the average deformation of the elastic bristle on the contact surface, v is the lifting velocity, σ_0 is the bristle stiffness coefficient, σ_1 is the bristle damping coefficient, and σ_2 is the viscous friction coefficient.

However, the model only takes into account the effect of speed, which is difficult to reflect the complex friction characteristics of the servo system at SCS low speed lifting. In order to overcome this deficiency, in this article, an improved LuGre model is constructed considering the specific characteristics of the SCS lifting servo system as shown in Fig. 4(b).

The mass of the SCS rod is gradually increased during the lifting process. Ideally, the mass change of the SCS $m_z(v)$ can be calculated by

$$m_z(v) = \pi d^2 h \rho / 4 = \pi v t d^2 \rho / 4 \quad (3)$$

where v is the lifting speed, t is the lifting time, d is the diameter of SCS rod, and ρ is the density of SCS rod. The total weight of the SCS rod W is then achieved by adding the $m_z(v)g$ to $m_w g$ at shoulder stage

$$W = (m_z(v) + m_w) g \quad (4)$$

where g is the acceleration of gravity.

In practice, due to the speed jitter and creeping phenomenon caused by nonlinear interference, the weight change of SCS is hard to be accurately calculated by the above equation. Therefore, a new partial differential equation is proposed in

this article to describe the law of weight change of SCS in the process of low-speed lifting, as follows:

$$\frac{\partial W}{\partial t} + v \frac{\partial W}{\partial v} = A v \rho g \quad (5)$$

where A is the cross-sectional area of the SCS rod.

In addition, the SCS lifting servo system is accompanied by a rotational motion during the low-speed lifting process. Rotation helps to evenly distribute the temperature around the crystal, reducing localized thermal stresses and thus avoiding defects within the crystal. At the same time, rotation induces the silicon atoms in the melt to be uniformly deposited on the surface of the crystals, which ensures the stability and uniform growth of the crystals [3]. In addition, rotation can also reduce the deposition of impurities on the crystal surface and inhibit compositional segregation, thus further improving the purity and structural integrity of produced SCS [32]. Therefore, the rotation process plays an indispensable and critical role in improving the quality of SCS. In this article, the impact of the SCS rotation is captured by multiplying the rotation speed with the SCS weight, based on which the coupling torque T_r can then be calculated by

$$T_r = K \omega_z \hat{W} \quad (6)$$

where K is the gain coefficient, ω_z is the SCS rotation speed, and $\hat{W}$ is the estimated weight of the SCS rod obtained by solving (5).

After the calculation of the weight variation in the SCS lifting servo system and the coupling torque generated by the SCS rotation, a torque influence factor σ_3 is defined to measure the influence of these two characteristics on the system friction. Hence, by adding this influence, the improved LuGre friction model is achieved

$$T_f = \sigma_0 z + \sigma_1 \frac{dz}{dt} + \sigma_2 v + \sigma_3 T_r \quad (7)$$

with

$$\begin{cases} \frac{dz}{dt} = v - \frac{|v|}{g(v)} z \\ g(v) = \frac{F_c + (F_s - F_c) e^{-\left(\frac{v}{v_s}\right)^2}}{\sigma_0} \end{cases} \quad (8)$$

where T_f is friction torque, F_c is the Coulomb friction, F_s is the static friction, and v_s is the Stribeck velocity.

Remark 1: The LuGre model is a widely used physical model for describing friction in different industrial systems. In this article, an improved LuGre friction model is achieved by considering SCS lifting characteristic, which serves as one of the physical constraints for friction prediction in the following section. The identification of the parameters in the physics-based LuGre model is an important research topic in industrial application. In the following section, the parameters can also be learned by the hybrid PINN strategy.

IV. FRICTION PREDICTION BY HYBRID LEARNING STRATEGY

In this section, the hybrid learning strategy for prediction of the friction in SCS lifting servo system is described, which fully takes into account the physical information presented in Section III and experiment data.

A. WE-PINN

The weight of the SCS rod is a key variable in the modeling of the friction as depicted in Fig. 3(b). In traditional friction prediction, the weight of the load is usually obtained by empirical formula or static measurement. However, empirical formula is difficult to adapt to dynamically changing operating conditions. Static measurement cannot accurately capture the instantaneous weight changes, especially for the complex operation of the SCS lifting servo system, where it is difficult to accurately measure the changes in the weight of the SCS rods due to the structural constraints of the crystal growth furnace. Generally, traditional methods lack a deep understanding of the complex nonlinear behavior in the dynamic changes of the system, resulting in poor accuracy of the prediction results.

To address these issues, a WE-PINN module is constructed to accurately estimate the SCS weight W . The module combines the powerful expressive power of neural networks with the physical constraints of PDE to ensure that the computation of W strictly follows the laws of physics and is able to respond to dynamic changes. In this way, WE-PINN not only improves the accuracy of weight calculation, but also enhances the model's ability to capture complex nonlinear behaviors.

In training of the WE-PINN, the PDE of weight is regarded as a physical constraint, by integrating the loss $L_{PDE} = (\partial W / \partial t) + v(\partial W / \partial v) - Avg$ into the loss function. Since the mass at the shoulder stage is hard to be accurately calculated, in this article, the weight change considered is in the equal-diameter stage. While the mass in the shoulder stage is regarded as an initial boundary condition constraint L_{IC} , which is described by $W(t=0, v) = m_w g$, where $W(t=0, v)$ denotes the initial SCS weight at $t=0$ of the equal-diameter stage. The inputs of the WE-PINN are the running time t and the SCS lifting speed v , which are used to estimate the real-time weight distribution during the crystal growth process.

B. FP-PINN

After obtaining the estimated SCS weight, the T_r in (6) is achieved, thus can be used for the friction model in (7). Then, the friction can be predicted by FP-PINN using the physics-based and data-based constraints, as depicted in Fig. 3(c).

For prediction of friction torque, the measurement data acquired from the experimental platform is needed. In this article, the measured data mainly include the SCS lifting speed v , SCS rotation speed ω_z , running time t , real-time current i of the motor, and actual friction torque T_{fm} .

With measurements and predefined neural network configuration, a loss function should be designed for network training. Before presenting the loss-function equation, we first compute several relevant parameters from the measured data.

On one hand, as described in Section III-A, the friction constraint to the dynamics of the servo system, T_{fpd} , can be obtained by

$$T_{fpd} = T_e - J \frac{d\omega_r}{dt} - B\omega_r \quad (9)$$

where the output torque T_e can be estimated based on the measured real-time current i of the motor by $T_e = K_t i$, with

K_t the torque constant. The corresponding motor rotational speed can be calculated from the measured lifting speed v by $\omega_r = (v/R)$, where R is the constant conversion factor. J and B are constants.

On the other hand, as described in Section III-B, the friction constraint to the improved LuGre friction model, T_{fpf} , can be expressed by

$$T_{fpf} = \sigma_0 z + \sigma_1 \frac{dz}{dt} + \sigma_2 v + \sigma_3 T_r \quad (10)$$

where T_r is calculated by (6) using the measured ω_z and estimated SCS weight $\hat{W}$ in Section IV-A. The terms (dz/dt) and z can be indirectly determined using (8) with measured v . While the remaining parameters $(\sigma_0, \sigma_1, \sigma_2, \sigma_3, F_s, F_c, v_s)$ in the improved LuGre friction model are regarded as learnable parameters. This friction term can reflect the friction characteristics during the ultra-low-speed SCS lifting process.

Based on the above terms, a combined loss function is then designed by considering three parts: the data loss L_{data} in the data-driven part, the dynamic loss L_{dyn} computed by considering the physical constraint of the dynamics model of the servo system, and the model loss L_{fr} computed with the physical constraint of the improved LuGre friction model. Using the mean square error three loss terms can be respectively formulated by

$$L_{data} = \frac{1}{N} \sum_{i=1}^N (T_{fp} - T_{fm})^2 \quad (11)$$

$$L_{dyn} = \frac{1}{N} \sum_{i=1}^N (T_{fpd} - T_{fp})^2 \quad (12)$$

$$L_{fr} = \frac{1}{N} \sum_{i=1}^N (T_{fpf} - T_{fm})^2 \quad (13)$$

where T_{fp} is the network-predicted friction torque, and T_{fm} is the actual friction torque. T_{fpd} and T_{fpf} are described in (9) and (10), respectively.

Then, the total loss L_{total} is computed by the sum of one data-driven loss and two physical loss

$$L_{total} = L_{data} + \lambda (L_{dyn} + L_{fr}) \quad (14)$$

where λ is physical loss weight. It worth noting that the loss term L_{data} enables the network to learn to match measured friction, L_{dyn} ensures predicted friction satisfies dynamic laws, and L_{fr} makes the LuGre model also conform to measured friction. They work together to realize a physics-informed data-driven friction torque prediction method. By this means, the servo dynamic model and the improved LuGre friction model are jointly involved in the iterative training to form the FP-PINN modeling network, where the neural network predicted friction torque T_{fp} can be constrained by both physical models.

With the measurements and calculation method of the loss function, the network parameters and a subset of the improved LuGre model parameters can be optimized. The learnable network parameters mainly comprise the connection weights and biases, which are analogous to those in standard networks. While, the learnable LuGre model parameters include $\sigma_0, \sigma_1, \sigma_2, \sigma_3, F_s, F_c$, and v_s . The FP-PINN takes four key operational

parameters as inputs: running time t , SCS lifting speed v , the angular velocity of the system ω_r , and the SCS rotation speed ω_z , thereby providing the time and kinematic information required for accurate friction prediction under dynamic operating conditions.

Remark 2: In this article, a two-stage hybrid WE-PINN and FP-PINN framework is designed. One is for the estimation of SCS weight and another is for the friction prediction. The two-stage PINNs both take into account the physical constraints and the experiment data information. In addition, the estimated SCS weight is integrated into the training of FP-PINN model significantly improves the prediction accuracy and robustness of the method. Compared with the traditional identification method, the WE-PINN method in this article can better solve the SCS weight estimation problem constrained by partial differential equations in (5), even when only a small amount of low-quality data is available. Regarding the friction prediction problem, purely data-driven methods normally demand large volumes of high-quality data to achieve satisfactory results. In contrast, the proposed FP-PINN incorporates physical constraints during training, thereby ensuring the credibility of the predictions even when the collected data are limited.

Remark 3: The learnable LuGre model parameters $\sigma_0, \sigma_1, \sigma_2, \sigma_3, F_s, F_c$, and v_s are with physical meaning and different units, i.e., N/m, N · s/m, N · s/m, —, N, N, and m/s, respectively. To ensure physical nature, all parameters are enforced to be positive by exponential parameterization (i.e., $\sigma_0 = \exp(\hat{\sigma}_0)$, where $\hat{\sigma}_0$ is regarded as the network-learned variable). Hence, the actual learning parameters becomes $\hat{\sigma}_0, \hat{\sigma}_1, \hat{\sigma}_2, \hat{\sigma}_3, \hat{F}_s, \hat{F}_c$, and $\hat{v}_s$. Consequently, the final values of all other parameters are obtained by taking the exponential of these optimized values, except for F_s . As for F_s , since $F_s > F_c$ is a hard constraint that must be satisfied to align with physical reality, a difference-based approach is adopted. $F_s = F_c + \exp(\hat{F}_s)$ is constructed to guarantee that F_s always remains larger than F_c .

V. EXPERIMENTS RESULTS

A. Experimental Platform

In this section, the validation of the proposed method are carried out based on a SCS lifting servo experimental platform built in our laboratory, as shown in Fig. 5. The experimental platform consists of the SCS lifting system, the motion control system, and the data acquisition module. As shown in Fig. 5(a), two motors are used for lifting and rotation, respectively. The rope is attached to a rigid block, simulating the SCS rod. As shown in Fig. 5(b), the motion control system includes a human-computer interface (TPC1031KI), a PLC (AS218TX-A) and a servo driver (A25A100). The touch screen is responsible for real-time displaying the system status and interacting with the operation commands. The PLC serves as the logic controller to complete the signal processing and communication. The servo driver accurately executes the motor control.

In Fig. 5(c), the data acquisition module is depicted. The system utilizes MCGS software to collect and store operation data at a sampling frequency of 2 Hz. Since the input ports of

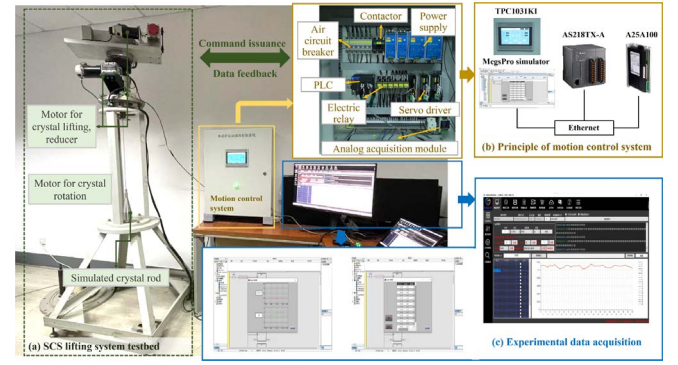


Fig. 5. SCS lifting servo system experimental platform.

PLC are fully occupied, the current signal is collected by an external third-party analog acquisition module. It connects the PLC with RS485 to collect the current monitoring signal of the servo driver.

B. Experimental Data Generation

The lifting speed of the SCS lifting servo system is usually in the range of [0.5, 2] mm/min. In order to cover more comprehensively the possible dynamic characteristics involved in the lifting process, the speeds are set to 0.5, 0.8, 1.2, 1.6, and 2 mm/min in the experiment. The rotational speed ω_z is set at 0.5 r/min. At each speed setting, several measurements are made and a complete set of motion data is collected. These data cover key physical quantities such as the SCS lifting speed v , SCS rotation speed ω_z , running time t , and real-time current i of the motor, and reflect the dynamic behavior of the SCS lifting servo system at different lifting speeds.

As for the actual friction torque data, due to the limitations of the experimental setup in this study, it currently is difficult to be measured directly. Hence an approximate approach is adopted. Specifically, the output torque data T_e are first calculated from the acquired measurements, and then the friction torque is approximated using (1) and taken as the actual value of friction torque. This calculation method is based on the assumption that during stable ultra-low speed lifting stage, other external forces are ignored, and only the impact of the change in the weight of the SCS rod is considered, which is reflected in the measured T_e . Although this approximation introduces some inaccuracy, it does not affect the validation of the proposed method. In addition, the key physical constants utilized in these computations and throughout the modeling framework, such as R and K_t , are summarized in Table I along with their values, units, and sources.

In order to improve the generalization ability and robustness of the modeling, the collected dataset is randomly divided into a training set and a test set, where the training set accounts for 70% and the test set accounts for 30%. Such a division ensures the diversity of data in the process of training the model, and at the same time retains enough test data for evaluating the performance and prediction accuracy of the model.

TABLE I
KEY PARAMETERS OF THE SERVO SYSTEM

Parameter	Description	Value	Unit	Source
J	Equivalent rotational inertia	2.7	$\text{kg} \cdot \text{m}^2$	Motor datasheet
B	Viscous damping coefficient	0.27	$\text{N} \cdot \text{m} \cdot \text{s}/\text{rad}$	Motor datasheet
R	Conversion factor	6.28	mm/r	Motor datasheet
K_t	Torque constant	0.227	$\text{N} \cdot \text{m}/\text{A}$	Motor datasheet
A	Cross-sectional area of the SCS rod	$\pi d^2/4$	mm^2	$A = \pi d^2/4$, where d is the rod diameter

TABLE II
PINN CONFIGURATION AND HYPERPARAMETERS SETTING

Parameter	Value	Parameter	Value
Layer size	5	Physical loss weight	1
Batch size	64	Dropout rate	0.3
Initial learning rate	$8e-4$	Activate function	ReLU
Epoch	$1e4$	Adaptive optimizer type	AdamW

C. Experimental Results of the Friction Prediction

After generating the dataset, the proposed FP-PINN and WE-PINN are trained according to the strategy described in Section IV. The construction and training of the network is done based on Python and PyTorch. The PINN has five hidden layers, where the number of nodes decreases layer by layer. The training process uses an adaptive optimizer, AdamW, combined with a customized loss function to ensure that the prediction results not only conform to the experimental data but also follow the physical constraints.

The hyperparameters of the PINN mainly includes layer size, batch size, initial learning rate, and number of epoch. These hyperparameters are selected through iterative trial-and-error to balance model accuracy against training efficiency. For instance, the layer size affects model complexity and accuracy, while the learning rate influences training efficiency. The PINN configuration and the hyperparameters settings are summarized in Table II.

To evaluate the performance of the proposed method, several methods are used for comparison, including the traditional parameter identification method using least squares recursion (LSR) to estimate the weight and FP-PINN with improved LuGre model (abbreviated as LSR-ILG), the WE-PINN and neural network method without consideration of physical information (abbreviated as NN-NoPhys), the WE-PINN and neural network with improved LuGre model (abbreviated as NN-ILG), the WE-PINN and FP-PINN with the traditional LuGre model (abbreviated as PINN-TLG), and the proposed WE-PINN and FP-PINN with the improved LuGre model (abbreviated as PINN-ILG).

Figs. 6 and 7 present friction prediction results and the prediction residuals for five methods, respectively, at the lifting speed of 0.5 mm/min. The proposed hybrid PINN-ILG aligns most closely with actual value over the entire time horizon, outperforming the LSR-ILG, NN-NoPhys, NN-ILG, and PINN-TLG methods as depicted in Fig. 6. It can be observed from Fig. 7, the accuracy of the LSR-ILG method is acceptable since it uses the improved LuGre model. But, it cannot sustain precise

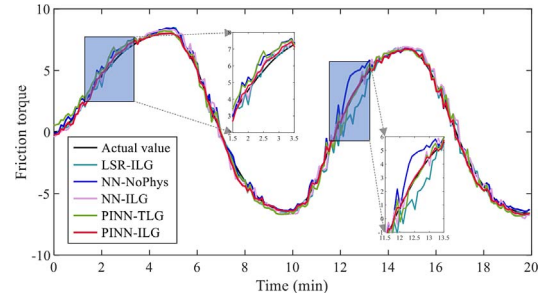


Fig. 6. Friction prediction results for different methods.

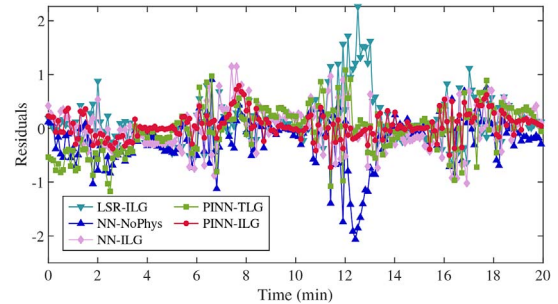


Fig. 7. Prediction residual of different methods.

predictions over the entire time horizon, because its adaptability still lags behind that of learning-based approaches. The NN-NoPhys method exhibits large fluctuations within certain time intervals. This occurs because the method omits physical constraints, rendering it sensitive to noise and unable to generalize beyond training conditions. The NN-ILG method partially mitigates these issues but still shows larger residuals compared to the proposed method. The PINN-TLG method exhibits systematic bias due to its static parameterization, indicating that it cannot adapt to the crystal growth characteristic including the SCS weight variation and the rotation of the SCS. The proposed PINN-ILG method yields the smallest residuals, demonstrating its accuracy and robustness in friction prediction during ultra-low-speed lifting process. This advantage is stemming from integrating physical friction mechanisms with data-driven learning.

Fig. 8 illustrates the training loss curves of the five methods across epochs. The proposed PINN-ILG method achieves rapid convergence with sustained low residual loss, benefiting from integrated physical constraints. The PINN-TLG method stabilizes at higher steady-state loss, as its physical model struggles

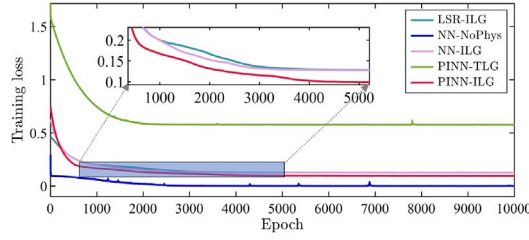


Fig. 8. Training loss curve of different methods.

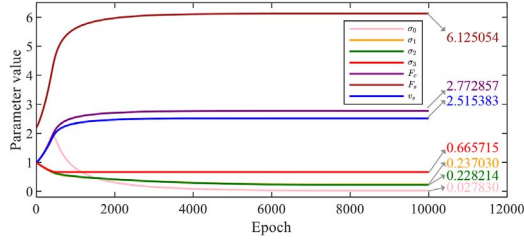


Fig. 9. Identification results for parameters in improved LuGre model.

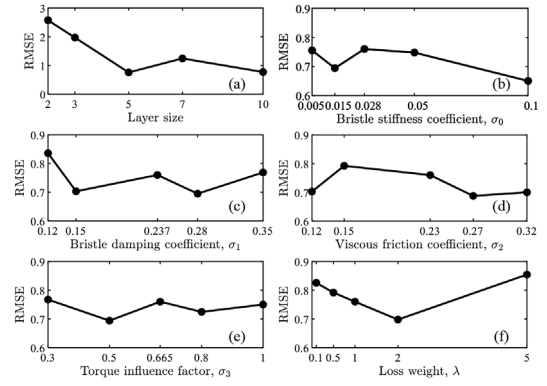
with complex friction dynamics. The NN-ILG method converges faster than the PINN-TLG method but lags the proposed method in final loss. The LSR-ILG method shows slower convergence and higher residual loss, reflecting limited flexibility. Meanwhile, although the NN-NoPhys method exhibits rapid convergence and achieves a low training loss, its lack of an explicit physical mechanism limits its applicability in practical engineering.

Fig. 9 illustrates the convergence of learned parameters in the improved LuGre model. After about 4000 epochs, all parameters stabilize. It can be observed that during the training process, the parameters are all positive and satisfy that the constraint $F_s > F_c$, as described in Remark 3 in Section IV-B. The final values for different parameters are color-coded in the figure. This demonstrates that iterative learning effectively optimizes physically interpretable parameters, confirming the model's self-adaptability to dynamic friction behavior.

D. Key Parameter Analysis

In this section, the key parameters in the proposed method are analyzed to evaluate the performance of the method. First, the impact of one of the key hyperparameters in PINN is discussed on the performance of proposed method, i.e., layer size. By varying the layer size to 2, 3, 5, 7, and 10, different friction torque prediction results as well the residual can be achieved. Recording the root-mean-square error (RMSE) of the residual, Fig. 10(a) is obtained. The result shows that the proposed method achieves the minimum RMSE, i.e., the best prediction accuracy, when the layer size is set to 5.

To further validate the stability and robustness of the proposed method under parameter variations, the sensitivity analysis about LuGre model parameters and physical loss weight are conducted. The LuGre model parameters mainly includes bristle stiffness coefficient σ_0 , bristle damping coefficient σ_1 ,


 Fig. 10. Key parameter analysis. (a) Impact of layer size on performance. (b)–(f) Sensitivity analysis of σ_0 , σ_1 , σ_2 , σ_3 , and λ .

viscous friction coefficient σ_2 , torque influence factor, σ_3 . Fig. 10(b)–(f) presents sensitivity analysis results. Taking Fig. 10(b) as an example, the sensitivity-analysis procedure in this section is illustrated below. First, the parameters σ_1 , σ_2 , σ_3 , and λ are fixed at 0.237, 0.23, 0.665, and 1, respectively. The parameter σ_0 is then varied individually across the values [0.005, 0.015, 0.028, 0.05, 0.1]. For each of these five settings, the proposed friction prediction PINN is employed to predict the friction torque, and the RMSE of the residuals is recorded, yielding Figs. 10(b). Sensitivity analyses for the remaining parameters follow the same procedure, except that parameter σ_0 is now held fixed at 0.028, yielding Figs. 10(c)–(f). It can be observed that although the prediction accuracy fluctuates slightly across the different settings, the variation remains small. This demonstrates that the proposed method is insensitive to parameter perturbations and thus exhibits good robustness.

E. Results Under Different Lifting Speeds

To evaluate the effectiveness of the proposed method under different working conditions, in this section, the experiment is conducted using different data obtained at different lifting speeds including (0.5, 0.8, 1.2, 1.6, 2) mm/min. Fig. 11 presents the hysteresis curve, which describes the relationship between the friction torque with respect to the speed. The hysteresis curve is commonly used for friction characteristic analysis. In Fig. 11, the green dots are the experiment data in different speeds, and the solid lines are the fitting results. It can be observed that the curves at different speeds match typical friction characteristics. At low speeds, friction torque is dominated by static friction. As speed increases, Coulomb friction gradually takes over, giving rise to the Stribeck curve. At high speeds, friction torque is governed by Coulomb friction and increases linearly with speed.

Fig. 12 presents the responses for position under a very small constant output torque. The output torque T_e is specifically set to 1.15, 1.10, and 1.05 times the static friction torque T_s , respectively. As shown in Fig. 12, the position response of the system exhibits distinct damped oscillatory characteristics. This oscillatory behavior confirms that the ILG model successfully

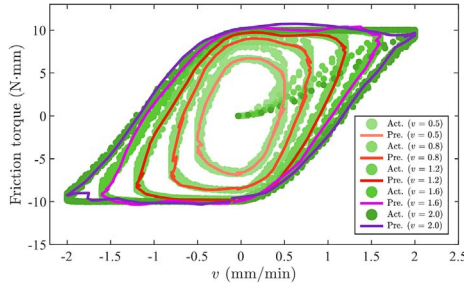


Fig. 11. Friction torque prediction results of SCS lifting servo system.

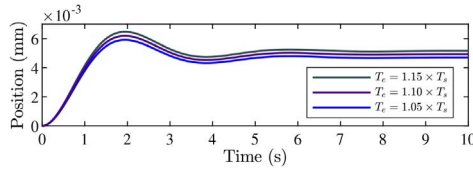


Fig. 12. Position output time response under very small constant torque.

captures static behavior of the friction dynamics, which similar to a second-order mass-spring-damper system.

To demonstrate the advantages of the proposed method, the remaining four methods mentioned in Section V-C are also performed at different lifting speeds (0.5, 0.8, 1.2, 1.6, 2) mm/min for comparison. All methods are trained multiple times with the hyperparameters kept the same. The root-mean-square error (RMSE) of predicted friction torque for each method is calculated. The RMSE values are illustrated in Fig. 13 and Table III. In Fig. 13, it can be seen that the proposed method outperforms the other four methods across most experimental conditions and maintains the lowest prediction error. Although the LSR-ILG, NN-NoPhys, and NN-ILG methods exhibit relatively stable prediction accuracy under different conditions, their RMSE values remain larger than the proposed method. The RMSE results of the PINN-TLG method show larger error fluctuations. Its performance at 1.6 and 2 mm/min approaches that of the proposed method, but its predictions deteriorate significantly at low speeds, indicating its difficulty in adapting to the varying working conditions of the crystal growth process. The NN-ILG, by fusing neural network adaptability with the LuGre model's physical interpretability, achieves more stable RMSE performance compared to the PINN-TLG, especially in the medium-to-high speed range (1.2 – 2.0 mm/min), though it still falls short of the proposed method's accuracy. The LSR-ILG method, despite nearing the proposed method's RMSE at the highest speed (2.0 mm/min), suffers from notably higher errors at low speeds (0.5 – 1.2 mm/min), reflecting its limited robustness across the full operating range.

To further comprehensively evaluate the prediction performance, the maximum absolute error (MAE) and R-squared (R^2) are additionally introduced for analysis. Fig. 14 illustrates the MAE value under different lifting speeds for five methods. The proposed PINN-ILG method maintains the lowest mean of MAE, while other four methods exhibit significant fluctuations

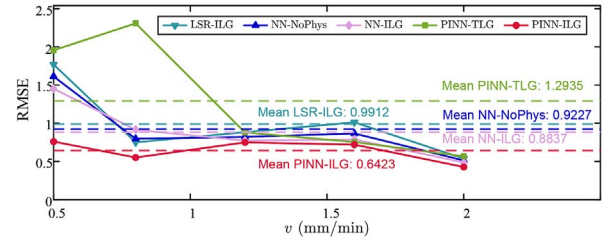


Fig. 13. RMSE for different methods at different speeds.

TABLE III
RMSE VALUES FOR DIFFERENT METHODS AT DIFFERENT SPEEDS

No.	$v = 0.5$	$v = 0.8$	$v = 1.2$	$v = 1.6$	$v = 2.0$	Average
1	1.7698	0.7510	0.8838	1.0129	0.5387	0.9912
2	1.6135	0.7973	0.8216	0.8669	0.5142	0.9227
3	1.4527	0.9161	0.7723	0.7889	0.4885	0.8837
4	1.9551	2.3080	0.8802	0.7558	0.5683	1.2935
5	0.7603	0.5519	0.7501	0.7209	0.4285	0.6423

Note: 1: LSR-ILG, 2: NN-NoPhys, 3: NN-ILG, 4: PINN-TLG, 5: PINN-ILG. The bold entries indicate the best performance achieved at different working conditions. Since smaller values represent better performance in this evaluation framework, the bold formatting highlights which method achieved the optimal (lowest) result.

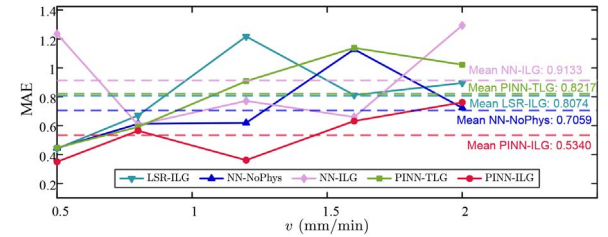


Fig. 14. MAE for different methods at different speeds.

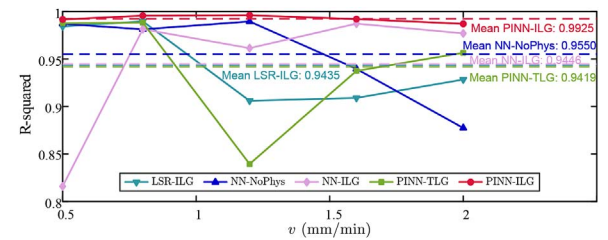


Fig. 15. R-squared for different methods at different speeds.

and with larger MAE. Fig. 15 presents the R^2 results. The proposed method consistently approaches 1, indicating strong explanatory power for friction torque variation. In contrast, the LSR-ILG and PINN-TLG methods show relatively lower R^2 at high speeds, and other methods also fall short of the proposed method. This demonstrates the good performance of proposed method across different working conditions.

Overall, the proposed method outperforms the other four methods in terms of prediction accuracy and robustness, proving its superior performance in friction torque prediction of the SCS lifting servo systems.

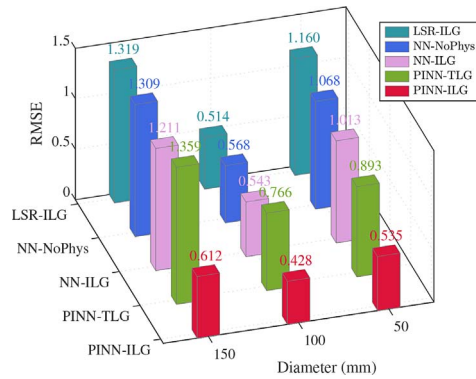


Fig. 16. RMSE of different methods for various SCS rod diameters.

F. Results Under Different Crystal Diameters

To further validate the effectiveness of the proposed method, generalization experiments under different diameters $d = (50, 100, 150)$ mm are conducted. The RMSE results of predicted friction in the SCS lifting servo system under three different sizes of SCS rods are shown in Fig. 16. The RMSE values of the proposed method (labeled by the red bar) under different diameters are all smaller than other four methods. The proposed method shows superior friction prediction accuracy. Hence, it can be concluded that the generalization ability of the proposed method under different sizes of SCS rods is enhanced through the improvement of the LuGre model and the integration of the two-stage PINN prediction strategy.

VI. CONCLUSION

In this article, a physics-informed data-driven method is proposed for modeling the friction dynamics in a SCS lifting servo system. By modifying the traditional LuGre friction model according to the characteristics of crystal growth and integrating the WE-PINN and FP-PINN learning strategy, the proposed method is effective in SCS lifting systems. Comparison with existing methods and generalization experiments show that the method outperforms other state-of-the-art techniques in terms of modeling accuracy and exhibits excellent generalization capability. In future research, friction compensation through high-accuracy prediction and modeling can be considered to support high-performance control of SCS lifting servo systems. For practical friction compensation applications, the prediction delay is also a crucial factor to be taken into account, which will be explored in our future work.

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